Bundled Fonts and Default-Typeface Changes in Major Operating Systems, Office Suites, and Document Tooling:

A Provenance Timeline Reference

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July 24, 2026
(working paper)

Abstract

The set of fonts bundled with an operating system or application, and the *default* typeface each release binds to its user interface and to new documents, change at datable public releases. Those dates bound the earliest time an artifact could have acquired the font through ordinary public channels (pre-release access—foundries, beta programs, OEMs—is the caveat), and default-font choices identify candidate producing software. This paper assembles verified timelines of (1) default system/UI font changes and bundled-font milestones in Windows, macOS, iOS, Android, and desktop Linux, including per-script CJK lanes (Simplified Chinese, Traditional Chinese, Hong Kong, Japanese, Korean); (2) default document-font changes in major office suites (Microsoft Office including its East Asian editions, OpenOffice.org/LibreOffice, WPS Office, Google Docs, Apple iWork, JustSystems Ichitaro, Hancom Office); (3) default and substitution fonts in PDF and scan/OCR tooling (Adobe Acrobat/Reader, PaperPort and OCR peers); and (4) default-font and font-substitution behavior in document-generation libraries (the Aspose suites and peers). Events are cited to the most authoritative sources located, with hotlinked URLs, and citations are annotated with verification tiers; where only secondary or incomplete sourcing was found, the citation and the claim say so, and claims that could not be adequately verified are labeled UNCERTAIN.

1 Introduction

A font cannot appear in a document before the font is available. The canonical demonstration is the Calibri affair of 2017 ("Fontgate"), in which property documents represented as executed in **February 2006** were typeset in Calibri—a font that first became publicly obtainable with Office 2007 Beta 2 on 2006-05-23 and generally available with Windows Vista and Office 2007 on 2007-01-30. The February 2006 dating is the decisive detail: the documents predate even the public beta, which featured in Pakistan's Supreme Court proceedings against the Sharif family.[1] Beyond first-availability bounds, *default* fonts carry a second, weaker signal: they identify the factory configuration of candidate producing software. This reasoning is asymmetric. A font used before its earliest public availability is decisive (an anachronism); a default-font match is only *consistent with* a toolchain, never proof of it, because users change defaults, templates override them, and fonts travel. Read this way: a PDF whose text layer is set in Fanwood, accompanied by Aspose's characteristic substitution warning or metadata, is strong evidence of Aspose.Words rendering on a font-starved Linux host (§6); text in Adobe Sans MM is consistent with Acrobat rendering a PDF whose fonts were missing (§5); a Simplified-Chinese document body in 等线 (DengXian) cannot predate 2015-07-29 (the font's first ship, in Windows 10) and is consistent with the Office 2016-era Simplified-Chinese default path (§4.1).

This paper is a dated, sourced reference for such reasoning. Each platform or product gets an Aeon-Timeline-style lane view plus an event register with citations.

2 Method and Evidence Tiers

Search protocol. Research was conducted in July 2026 by parallel per-platform investigations, each proceeding: (1) general web search to locate candidate sources for each event; (2) preference-ordered source selection—vendor primary documentation (Microsoft Learn/Typography, Apple Newsroom/Support/Developer, Android Developers Blog, project release notes and bug trackers, vendor API references), then institutional records (Computer History Museum, SEC EDGAR), then first-person designer accounts (Folklore.org; Ken Lunde's CJKV writings treated as near-primary), then reputable secondary/encyclopedic sources; (3) fetch-and-confirm verification of each cited URL against the specific claim (a 200 response serving unrelated content does not count as verification); (4) Wayback Machine fallback for dead originals. Searches used general-purpose web search engines plus direct navigation of the vendor documentation trees named above; fetching used both plain HTTP clients and rendering fetchers (several vendor sites are JavaScript-routed or bot-hostile, noted per citation). Inclusion required that a source bear directly on the dated event; user-forum material was admitted only for vendor-staff answers or where no better source exists, and is always tiered [secondary]. Artifact inspections (e.g., §4.1's theme dump) are documented with path, software version, observation date, and SHA-256. Conflicting sources are footnoted with both readings and a stated preference (vendor primary over encyclopedic—e.g., the Trebuchet MS first-ship discrepancy in §3.1). Negative findings ("no change found") are scoped to the sources actually searched and say so. The research window bounds all "current" statements. This protocol supports spot-checking every claim from its citation; it is not a systematic-review protocol, and §8 says so.

What counts as a font "appearing". Provenance reasoning must distinguish: (a) a *nominal reference*—a font name in document markup (a `.docx` run property, a PDF `/BaseFont` name) with no font data; (b) an *embedded font program*; (c) *rendered output*—glyph outlines actually drawn, possibly by a substituted face; (d) a *substituted display face* (the viewer's stand-in, §5); and (e) an *OCR text layer's* styling, which reflects the OCR engine, not the source document (§5.2). Anachronism logic applies strictly to (b)—an embedded font program cannot predate the font. For (a), a nominal reference is strong but rebuttable: font *names* can be written by software that never possessed the font (template inheritance, format converters, substitution tables—e.g., 2007-era iPhone pages could *request* Helvetica Neue years before the font shipped on the device, §3.5), so a nominal anachronism obliges the examiner to rule out name-injection paths before concluding. Inferences from (c)–(e) concern the *rendering/processing* toolchain, not the original author. The timelines below date public availability; each anchor's type is stated in §7.

Evidence tiers. Citations below carry one of these statuses, assigned during research in July 2026:

- [**verified live**]—the URL was fetched during this research and its content was confirmed to support the claim (an HTTP 200 serving an error or unrelated page does not qualify).
- [**archive**]—the original is dead or unreachable; an Internet Archive snapshot is cited. Where the snapshot's *existence* was confirmed via the Wayback availability API but its body could not be re-read during this research (archive.org rate-limiting), the citation says so explicitly.
- [**secondary**]—a reputable non-primary source (encyclopedic or press), used for corroboration or where no primary survives; presented as such.
- **UNCERTAIN**—the claim is plausible and commonly reported but could not be verified against an adequate source during this research; it is flagged, not asserted.

Primary sources are preferred throughout: vendor documentation (Microsoft Learn/Typography font-family pages, Apple Support "Fonts included with…" articles, Apple Newsroom press releases, Android Developers Blog, GNOME/KDE/Fedora/Debian/Ubuntu project records, Adobe HelpX and the PDF/ISO specification, vendor API references), institutional records (Computer History Museum), and first-person designer accounts (Folklore.org; Ken Lunde's CJK writings are treated as near-primary for CJKV matters).

Script lanes, not retail regional editions. For CJK coverage this paper models *script/locale lanes*—Simplified Chinese (SC), Traditional Chinese (TC), Traditional Chinese–Hong Kong (TC-HK), Japanese (JP), Korean (KR)—rather than retail regional editions (a "Taiwan edition" vs a "mainland edition"). Three facts justify this choice. First, the Unicode consolidation era decoupled CJK font availability from retail regional editions: Windows 2000 carried the East Asian fonts on every edition's installation media (installable via Regional Options—though pre-Vista Windows still shipped single-language OS images, with MUI an add-on), and Mac OS X and the mobile OSes bundled multi-script font sets from their first releases; what varies by locale is primarily the per-locale default UI font binding and, in office software, the per-language document defaults—not whether the CJK fonts can be present.[2, 3] Second, regional editions matter as distinct artifacts only in the pre-Unicode era (Chinese/Japanese/Korean Windows 3.x/9x; KanjiTalk-era classic Mac OS), and those appear here as dated events *within* the script lanes. Third, Singapore has no distinct font lane—zh-SG follows Simplified Chinese conventions—while Hong Kong earns a sub-lane solely because of the HKSCS character-set supplements and, later, dedicated HK typefaces (PingFang HK, §3.4).[4]

A note on Windows 2000 specifically, because it anchors the "editions stop mattering" argument: on Windows 2000 and XP the East Asian fonts shipped on the installation media of every edition but were **not installed by default** outside CJK locales (Regional Options → "Install files for East Asian languages"); CJK fonts became installed-by-default regardless of locale only in the Windows Vista/7 era.[5]

3 Operating Systems

3.1 Microsoft Windows — Latin lane

Figure 1 traces the Windows default UI font and bundled-font milestones.

Event register.

- **1985-11-20 — Windows 1.0.** UI set in bitmap fonts: a proportional "System" font for titles/menus and "Helv" for dialogs; no scalable outlines.[6]

1985	Windows 1.0 (Nov 20)—bitmap System font (titles and menus) with Helv bitmap for dialogs
1992	Windows 3.1 (Apr 6)—TrueType arrives with Arial, Times New Roman, Courier New; Helv renamed MS Sans Serif
1995	Windows 95 (Aug 24)—MS Sans Serif as default UI face; Comic Sans MS via Plus! pack
1998	Windows 98 (Jun 25)—Verdana, Tahoma, Trebuchet MS present per Microsoft font tables
2000	Windows 2000 (Feb 17)—Tahoma replaces MS Sans Serif as default UI font; Georgia bundled
2001	Windows XP (Oct 25)—Tahoma retained
2007	Windows Vista (Jan 30)—Segoe UI 9 pt becomes default; ClearType six (Calibri, Cambria, Candara, Consolas, Constantia, Corbel) ship
2012	Windows 8 (Oct 26)—Segoe UI stylistic refresh, Semilight weight
2015	Windows 10 (Jul 29)—Segoe UI retained; many script fonts move to optional Feature-on-Demand packs
2021	Windows 11 (Oct 5)—Segoe UI Variable becomes default; Cascadia Code and Cascadia Mono bundled
2023	Aptos (Jul 13)—replaces Calibri as Microsoft 365 default; distributed via Office cloud fonts, never bundled in Windows

Figure 1: Windows default UI font and bundled-font milestones (Latin lane).

- **1992-04-06 — Windows 3.1.** TrueType introduced, with the founding scalable core set **Arial, Times New Roman, Courier New** (plus Symbol)—the first Windows with scalable outline fonts. The "Helv" bitmap is renamed **MS Sans Serif**. Launch date per the Computer History Museum.[6–8]
- **1995-08-24 — Windows 95. MS Sans Serif** continues as the default UI font (through NT 4.0, 98, ME). **Comic Sans MS** first ships with the Windows 95 Plus! pack and OEM releases, per Microsoft's own font page.[6, 9, 10]
- **1998-06-25 — Windows 98.** Microsoft's per-font supply tables list **Verdana** (v2.10), **Tahoma** (v2.25), and **Trebuchet MS** (v0.98) from Windows 98. Verdana and Georgia (both Matthew Carter, 1996) reached users earlier via Internet Explorer and the 1996 "Core fonts for the Web" program; Wikipedia's typeface list attributes Verdana/Tahoma to Windows 95 via that route, while Microsoft's own tables enumerate only from 98—and for Trebuchet MS Microsoft's page (Windows 98) contradicts Wikipedia's "2000". Microsoft's pages are preferred here; the discrepancies are noted rather than resolved.[11–16]
- **2000-02-17 — Windows 2000. Tahoma replaces MS Sans Serif** as the default screen/UI font (through XP and Server 2003). **Georgia** appears from Windows 2000 in Microsoft's table.[17–19]
- **2001-10-25 — Windows XP.** Tahoma retained as default UI font.[18, 20]
- **2007-01-30 — Windows Vista (consumer GA; business availability 2006-11-30). Segoe UI at 9 pt becomes the default UI font**, per Microsoft's Win32 UX guidance. The **ClearType Font Collection**—Calibri, Cambria, Candara, Consolas, Constantia, Corbel (plus Cambria Math and Meiryo)—ships to the general public; it had first appeared publicly in Office 2007 Beta 2 on 2006-05-23. In Office 2007, released the same day, **Calibri replaced Times New Roman as the Word default** and Arial as the Excel/PowerPoint/Outlook default (§4.1). Provenance anchor: a document purportedly authored before mid-2006 set in Calibri is anachronistic.[21–24]
- **2009-10-22 — Windows 7.** Segoe UI 9 pt retained (Light/Semibold/Symbol weights added).[22, 25]
- **2012-10-26 — Windows 8.** Segoe UI ships a stylistic redraw (dated October 2011), with a Semilight weight; Black and Emoji additions follow in 8.1 (2013-10-17).[26, 27]
- **2015-07-29 — Windows 10.** Segoe UI retained. Structurally important for provenance: **many script-specific fonts move into optional Feature-on-Demand (FOD) packages**, installed automatically only when the associated language is enabled—presence of a font on a Windows 10/11 system no longer implies the OS installed it unconditionally.[28, 29]
- **2021-10-05 — Windows 11. Segoe UI Variable** (weight axis plus an optical-size axis) becomes the default UI face; the Windows 11 font list marks Segoe UI Variable, Segoe Fluent Icons, Cascadia

Code, and Cascadia Mono "Added in Windows 11".[30, 31]

- **2023-07-13 — Aptos announced as the Microsoft 365 default** (Steve Matteson's "Bierstadt" renamed), replacing Calibri across Word, Excel, PowerPoint, Outlook. Distribution boundary, precisely: Aptos reaches users (a) inside Microsoft 365 apps via the Office Font Service ("cloud fonts") and (b) as a free manual download from the Microsoft Download Center (v4.40 published 2024-06-04) for use outside Office; it is **not** bundled as a Windows operating-system font and is absent from Microsoft's Windows 11 font list.[30, 32–35]

Microsoft publishes authoritative per-version "Fonts that ship with Windows" lists for Windows 7 through 11; no such pages exist for Vista or XP (retired URLs return 404), for which Wikipedia's typeface list with its "Included from" column is the best available consolidated record.[16, 36]

3.2 Microsoft Windows — CJK script lanes

Figure 2 traces the Windows CJK default UI fonts by script lane.

Simplified Chinese (SC). The ZhongYi (Beijing ZhongYi Electronics) faces **SimSun 宋体** and **SimHei 黑体** anchor the lane from the regional Windows 3.x/95 editions (pre-2000 per-edition ship dates rest on secondary sources; Microsoft's supply tables begin at Windows 2000, where SimSun/NSimSun v2.11 first appear). SimSun was the **default SC UI font on Windows 2000/XP**. On **Vista (2007-01-30)**, **Microsoft YaHei 微软雅黑**—glyphs licensed from Beijing Founder Electronics, ClearType-optimized—became the default SC UI font; SimSun remained bundled and **SimSun-ExtB** (CJK Extension B) was added. **DengXian 等线** ships from Windows 10 as an optional Feature-on-Demand font (it is also Office's Chinese default from Office 2016, §4.1). Windows 11 retains YaHei UI as the SC UI binding, since Segoe UI Variable covers only Latin/Greek/Cyrillic.[37–42]

Traditional Chinese (TC). The DynaLab (later DynaComware, 華康) Ming face—raster in TC Windows 3.0, TrueType from 3.1, renamed **MingLiU 細明體** at v2.00 ("U" for Unicode)—was the TC UI face from Windows 3.0 through XP, joined by **DFKai-SB 標楷體** (kai) and, from TC Windows 98, the proportional **PMingLiU 新細明體** (default TC UI font on 2000/XP). On **Vista**, **Microsoft JhengHei 微軟正黑體** became the default TC UI font. Its designer is **China Type Design Limited** of Hong Kong (a Monotype company)—not DynaComware, a frequent misattribution. MingLiU-ExtB/PMingLiU-ExtB arrived with Vista; the whole legacy Ming/Kai set moved to Feature-on-Demand in Windows 10. JhengHei UI remains the TC binding in Windows 11.[40, 43–47]

Traditional Chinese — Hong Kong (HKSCS). Hong Kong's supplementary character sets (GCCS 1995; HKSCS-1999/-2001/-2004/-2008/-2016) reached Windows first as **patches** for Windows 98/NT4/2000/XP that enabled a hidden code page 951 and patched MingLiU. **Vista** brought native, Unicode-based **HKSCS-2004** support with dedicated **MingLiU_HKSCS** and **MingLiU_HKSCS-ExtB** fonts, which remain the bundled HK glyph carriers (v7.03, in the TC Feature-on-Demand set) through Windows 10/11.[48, 49]

Japanese (JP). **MS Gothic** was "designed by Ricoh Co. Ltd. and licensed as the default system font for Windows 3.1" (1993), with serif companion **MS Mincho**; **MS UI Gothic** (from v2.30, Windows 2000) carried the JP UI through XP. On **Vista**, **Meiryo メイリオ**—designed by C&G Inc. (Satoru Akamoto, Takeharu Suzuki, Yukiko Ueda) with **Eiichi Kōno** and **Matthew Carter**, JIS X 0213:2004-compliant, part of the ClearType collection—became the default JP system font; **Meiryo UI** followed in Windows 7. **Windows 8.1 (2013-10-17)** bundled Jiyukobo's **Yu Gothic 游ゴシック** and **Yu Mincho 游明朝** while keeping Meiryo UI as the UI face; the switch of the JP UI font to **Yu Gothic UI** came with **Windows 10 (2015-07-29)**, at which point Meiryo and MS Gothic/Mincho moved to the JP supplemental FOD set. **UD Digi Kyokasho** (universal-design textbook faces) arrived in Windows 10 version 1709.[50–54]

Korean (KR). HanYang I&C's **Gulim 굴림/GulimChe** (gothic) and **Batang 바탕/BatangChe** (myeongjo) anchor the lane from 1990s Korean Windows (Microsoft-documented from Windows 2000, v2.20); **Gulim** was the KR UI font on 2000/XP. On **Vista**, **Malgun Gothic 맑은 고딕**—Sandoll Communications (designers Kyoung-bae Lee and Daekwon Kim), Latin glyphs derived from Segoe UI—became the default KR UI font and has remained so through Windows 11 (hanja added in Windows 8 via China Type Design; Semilight weight later); Gulim/Batang/Dotum moved to the KR FOD set in Windows 10.[55–58]

3.3 Apple macOS — Latin lane

Figure 3 traces the classic Mac OS and macOS default system font.

Event register.

- **1984-01-24 — System 1 (Macintosh 128K).** Default system font **Chicago** (Susan Kare; 12 pt bitmap), used for menus, dialogs and titles, and retained as the system font through System 7.6. The original bundle is the bitmap "city" set—Chicago, Geneva, Monaco, New York, Venice, London, Athens, San Francisco (the 1984 ransom-note face, unrelated to the 2015 system font), 1993 Windows 3.1J—MS Gothic and MS Mincho by Ricoh licensed as the Japanese system fonts

1998 TC Windows 98—PMingLiU (New MingLiU) joins DynaLab MingLiU

2000 Windows 2000 (Feb 17)—East Asian fonts carried on every edition's install media (installable via Regional Options, not default-installed)—SimSun (SC), PMingLiU (TC), MS UI Gothic (JP), Gulim (KR) as locale UI fonts

2007 Windows Vista (Jan 30)—the four-lane ClearType switch—Microsoft YaHei (SC), Microsoft JhengHei (TC), Meiryo (JP), Malgun Gothic (KR); HKSCS-2004 native; ExtB fonts added

2013 Windows 8.1 (Oct 17)—Yu Gothic and Yu Mincho bundled (Jiyukobo); Meiryo UI still the JP UI font

2015 Windows 10 (Jul 29)—JP UI font becomes Yu Gothic UI; legacy CJK families move to Feature-on-Demand; DengXian ships as SC FOD font

2021 Windows 11 (Oct 5)—YaHei UI, JhengHei UI, Yu Gothic UI, Malgun Gothic all retained; Segoe UI Variable adds no CJK coverage

Figure 2: Windows CJK default UI fonts by script lane.

1984 System 1 (Jan 24)—Chicago by Susan Kare, with the bitmap city fonts (Geneva, Monaco, New York and others)

1991 System 7 (May 13)—TrueType introduced

1997 Mac OS 8 (Jul 26)—Charcoal (Font Bureau, 1995) replaces Chicago as default; system font user-selectable

2000 Mac OS X Public Beta (Sep 13)—Lucida Grande debuts with Aqua

2001 Mac OS X 10.0 (Mar 24)—Lucida Grande confirmed as the OS X UI font

2014 OS X 10.10 Yosemite (Oct 16)—Helvetica Neue replaces Lucida Grande

2015 OS X 10.11 El Capitan (Sep 30)—San Francisco replaces Helvetica Neue; Apple's first in-house system typeface in two decades

Figure 3: Classic Mac OS and macOS default system font (Latin lane).

Toronto, Cairo—largely Kare's work; her first-person account of the city-naming convention survives on Folklore.org. The launch date is the 1984-01-24 shareholders'-meeting unveiling, anchored institutionally by the Computer History Museum.[59–64]

- **1991-05-13 — System 7. TrueType** introduced—Apple's outline font format, later licensed to Microsoft—freeing the Mac from fixed-size bitmaps (a TrueType Chicago was cut to match the bitmap metrics). Date corroborated by AppleInsider's dated retrospective; contemporaneous 1991 primary not captured.[65–67]
- **1997-07-26 — Mac OS 8.** Default system font changes to **Charcoal** (designed 1995 at Font Bureau, based on Chicago), introduced with the Platinum appearance; the Appearance control panel made the system font user-selectable, with Chicago still available. The 1997-07-26 ship date rests on secondary sources; Apple's 1997 press release survives only in Wayback snapshots that could not be re-read during this research (archive.org rate limiting)—flagged accordingly.[68–71]
- **2000-09-13 / 2001-03-24 — Mac OS X Public Beta ("Kodiak"), then 10.0 ("Cheetah"). Lucida Grande** (Bigelow & Holmes' Lucida family) becomes the UI font of the Aqua interface, first in the Public Beta and then in 10.0, and remains so until 2014. Apple's 2001-03-21 press release "Mac OS X Hits Stores This Weekend" fixes the 10.0 retail date ("beginning this Saturday, March 24") and—a bundled-font primary in its own right—advertises "more than $1,000 of the best fonts", naming Baskerville, Zapfino, Futura and Optima, plus "the highest-quality Japanese fonts" (§3.4).[72–74]
- **2014-10-16 — OS X 10.10 Yosemite.** Default system typeface changes from Lucida Grande to **Helvetica Neue**, aligning the Mac with iOS; the GA date is fixed by Apple's press release. The font-switch attribution itself rests on the encyclopedic record and John Siracusa's Ars Technica review (the review's typography section could not be re-fetched during this research; Ars blocks automated readers).[75–77]
- **2015-09-30 — OS X 10.11 El Capitan.** Default system font changes from Helvetica Neue to **San Francisco**, Apple's first in-house system typeface since the 1980s bitmap era, introduced across plat-

forms (iOS 9: 2015-09-16, §3.4; watchOS earlier, 2015-04-24). Primary anchors: Apple's availability press release and the WWDC 2015 session "Introducing the New System Fonts" (Antonio Cavedoni).[78–80]

San Francisco family evolution (bundled-font milestones). SF Compact shipped first on Apple Watch (watchOS, April 2015); **SF Mono** appeared at WWDC 2016 (Terminal/Xcode); the downloadable family was renamed from "SF UI" to **SF Pro** in the WWDC 2017 timeframe—supported by the SF Pro font files' own copyright string ("© 2015–2017 Apple Inc.") and the documented naming history; the before/after Wayback snapshots of Apple's developer fonts page (2016-06-18 vs 2017-06-12) exist per the availability API but their bodies could not be re-read during this research, so the 2017 rebrand year is stated with that caveat. **New York**, the serif companion (first seen as "SF Serif" in Apple Books, iOS 12), was released on the Apple Developer site on **2019-06-03**. Variable-font builds of SF arrived at WWDC 2020, and the **SF script extensions** (SF Arabic at WWDC 2021, later SF Armenian, SF Georgian, SF Hebrew) extend the family beyond Latin.[80–83]

Bundled-font inventories and removals. Apple publishes a per-release "Fonts included with macOS <name>" support document (e.g., article 120414 for macOS Sequoia), distinguishing pre-installed fonts, download-on-demand fonts (dimmed in Font Book), and document-support-only fonts—the authoritative per-version inventory. Since the Sierra era many legacy families (including the classic TC faces, §3.4, and Osaka) are download-on-demand rather than pre-installed.[84]

3.4 Apple — CJK lanes (macOS and iOS)

Figure 4 traces the Apple CJK system fonts by script lane.

Japanese. The KanjiTalk-era system face was **Osaka** (bitmap; TrueType by the System 7/KanjiTalk 7 era—classic-era dates approximate). The modern era begins with Apple's **2000-02-16 Macworld Tokyo announcement**—still live on Apple Newsroom—of an agreement with Dainippon Screen (fonts designed by **Jiyukobo**) to bundle **Hiragino**: "six typefaces, 17,000 characters each, in OpenType", the JIS X 0213 expanded set. Hiragino shipped with Mac OS X 10.0 (2001-03-24), and the Hiragino gothic faces have been the Japanese UI lineage on Mac OS X since—initially as **Hiragino Kaku Gothic Pro**, represented since the El Capitan era by the reorganized **Hiragino Sans** family (weights W0–W9). Osaka persists as a download-on-demand legacy font.[85–87]

Simplified Chinese. Mac OS X bundled Changzhou SinoType's ST faces; **STHeiti became the SC UI font in 10.2 Jaguar** (2002)—not 10.0, a common conflation—with the full outline family (STXihei, STSong, STKai, STFangsong) present by 10.3. **Hiragino Sans GB** (an SC adaptation with Beijing Hanyi) was added in 10.6 Snow Leopard (2009-08-28). The ST families were regrouped as Heiti SC / Songti SC / Kaiti SC collections in the 10.7/10.8 era.[88, 89]

Traditional Chinese and Hong Kong. The Chinese Language Kit era (∼1995) brought DynaLab's **Apple LiSung 儷宋** and **Apple LiGothic 儷中黑** plus BiauKai 標楷. 10.3 Panther added **LiHei Pro 儷黑 Pro** and **LiSong Pro 儷宋 Pro**—Big-5E character set **including HKSCS**, making them the HK carriers—with LiHei Pro as the TC UI face through 2015. The legacy TC faces were demoted to download-on-demand around the Sierra era (not removed in Catalina, a common misreport).[87, 88]

The 2015 PingFang event. With iOS 9 (2015-09-16) and OS X 10.11 El Capitan (2015-09-30), **PingFang 蘋方** became the Chinese UI family—produced with DynaComware, six weights, in three regional variants: **PingFang SC, PingFang TC, and PingFang HK**. This is Apple's first dedicated Hong Kong UI lane (previously HK support rode inside Taiwan-TC fonts via HKSCS), and the event replaced STHeiti/Heiti SC/TC and LiHei Pro simultaneously.[90–92]

Korean. **AppleGothic** carried the KR UI from the Korean Language Kit era through 2011–2012, when Sandoll's **Apple SD Gothic Neo** replaced it—first on **iOS 5 (2011)**, then on **OS X 10.8 Mountain Lion (2012)** with nine weights; it remains bundled (macOS Sequoia ships v19.x). The iOS 5.0-vs-5.1 point and the Mountain Lion attribution rest on foundry/community records rather than an Apple statement—flagged accordingly.[93, 94]

iOS bundled-font history (Latin note in passing). The original iPhone (2007) shipped a curated subset of Mac faces—Gruber's contemporaneous PDF specimen, still live, classifies 9 full families installed (Arial, Courier New, Georgia, Helvetica, Marker Felt, Times New Roman, Trebuchet MS, Verdana, Zapfino), 2 partial, 3 name-substituted (**Helvetica Neue substituted by Helvetica**—Neue was not actually installed until the iPhone 4 era), and 27 Mac families absent (§3.5).[95, 96]

3.5 Apple iOS — Latin lane

Figure 5 traces the iOS default system font.

Event register.

- **2007-06-29 — iPhone OS 1.0.** System font **Helvetica**; Notes uses Marker Felt. Helvetica Neue was

1986	KanjiTalk era—Osaka as the Japanese system face
Mid-1990s	Chinese Language Kit (version dates approximate)—Apple LiSung and Apple LiGothic (DynaLab) for TC; Song, Hei, Kai, Beijing for SC
2000	Apple announces Hiragino licensing (Feb 16, Macworld Tokyo)—six OpenType faces, 17,000 characters each
2001	Mac OS X 10.0 (Mar 24)—Hiragino ships; Kaku Gothic Pro is the JP UI face
2002	Mac OS X 10.2 Jaguar—STHeiti (SinoType) becomes the SC UI font
2003	Mac OS X 10.3 Panther—LiHei Pro and LiSong Pro (DynaComware, Big-5E with HKSCS) join; LiHei Pro is the TC UI face
2009	Mac OS X 10.6 Snow Leopard (Aug 28)—Hiragino Sans GB added for SC
2011	iOS 5—Apple SD Gothic Neo (Sandoll) replaces AppleGothic as the KR UI font on iOS
2012	OS X 10.8 Mountain Lion—Apple SD Gothic Neo becomes the KR UI font on the Mac
2015	iOS 9 (Sep 16) and OS X 10.11 (Sep 30)—PingFang SC, TC and HK become the Chinese UI fonts; Hiragino rebranded Hiragino Sans

Figure 4: Apple CJK system fonts by script lane (classic Mac OS, macOS, iOS).

2007	iPhone OS 1.0 (Jun 29)—Helvetica as system font; Helvetica Neue not installed, only name-substituted
2010	iPhone 4 with iOS 4 (Jun 21 software, Jun 24 device)—Helvetica Neue on Retina hardware; non-Retina devices keep Helvetica
2013	iOS 7 (Sep 18)—flat redesign leans on Helvetica Neue Light and Ultralight; beta 3 walks small text back to Regular
2015	iOS 9 (Sep 16)—San Francisco replaces Helvetica Neue

Figure 5: iOS default system font.

not installed—pages requesting it silently got Helvetica (Gruber's specimen, §3.4). The full inventory: 9 full families, 2 partial, 3 substituted names, 27 absent.[95–97]

- **2010-06-21 / 2010-06-24 — iOS 4 and iPhone 4. Helvetica Neue becomes the system font on Retina hardware.** This change is device-driven, not an OS-version change: the same iOS 4 on non-Retina devices (3GS and earlier) kept Helvetica—Gruber's contemporaneous review is explicit, attributing the split to hinting quality on low-density screens. Apple's iPhone 4 press release carries both dates (iOS 4 free update June 21; iPhone 4 on sale June 24) and the Retina display.[98, 99]
- **2013-09-18 — iOS 7.** The flat redesign makes **Helvetica Neue** carry the interface system-wide, leaning on **Light/Ultralight** weights, with Dynamic Type introduced. In the beta cycle, betas 1–2 set most UI text in Helvetica Neue Light and were widely criticized; **beta 3 (2013-07-08) walked small text back to Regular**—a documented within-cycle default correction. GA date per Apple's press release.[100–102]
- **2015-09-16 — iOS 9. San Francisco** replaces Helvetica Neue as the system font, in lockstep with El Capitan (§3.3); SF had debuted on Apple Watch on 2015-04-24 and been available to developers since 2014-11-18. GA date per Apple's press release.[80, 103]
- **2019-09-19 — iOS 13 (provenance-relevant API change).** Third-party **custom font installation** becomes supported system-wide—fonts distributed through App Store apps, registered via `CTFontManagerRegisterFontURLs`, managed in Settings (WWDC 2019 session 227). After this point, presence of a non-Apple font on an iOS device no longer implies jailbreak or configuration-profile sideloading.[104–106]

3.6 Android

Figure 6 traces the Android default fonts (Latin and CJK).

Event register (Latin lane).

- **2007-11-12 — Droid family announced.** Ascender Corporation announces the Droid collection (Droid Sans, Droid Serif, Droid Sans Mono), designed by **Steve Matteson**, commissioned for the Open Handset Alliance's Android platform. The original PRWeb release is dead; a Wayback snapshot exists (existence confirmed via the availability API; body not re-read during this research due to archive.org rate limiting), with content corroborated by the encyclopedic record.[107, 108]
- **2008-09-23 — Android 1.0 (SDK release 1; the first shipping handset, the T-Mobile G1, reached customers in October 2008). Droid Sans** is the default UI font; **Droid Sans Fallback**—Apache-licensed, ~15,500 Han glyphs built

2007	Droid family announced (Nov 12)—Ascender Corporation, designer Steve Matteson, for the Open Handset Alliance
2008	Android 1.0 (SDK Sep 23, first handset Oct)—Droid Sans default UI, Droid Serif and Droid Sans Mono bundled, Droid Sans Fallback for CJK
2011	Android 4.0 Ice Cream Sandwich (Oct 19 SDK)—Roboto replaces Droid Sans; MotoyaLMaru and MotoyaLCedar bundled for Japanese
2014	Roboto v2 published (Jul 16) with Material Design; Noto Sans CJK released (Jul 15); both ship in Android 5.0 Lollipop (Nov 12)
2015	Android 6.0 Marshmallow—legacy Droid Sans Fallback and Motoya files removed, Noto covers all CJK
2017	Android 8.0 Oreo (Aug 21)—Downloadable Fonts API and fonts-in-XML; Google Sans appears on Pixel skin (8.1)
2022	Roboto Flex variable font released to Google Fonts (May 5)—not adopted as the AOSP system default

Figure 6: Android default fonts (Latin and CJK).

by component reference, following mainland-PRC preferred glyph shapes—is the single CJK fallback (the root of later "wrong-region glyph" complaints in TC/JP contexts). Primary date anchor: the Android Developers Blog post announcing the Android 1.0 SDK, release 1.[108–110]

- **2011-10-19 — Android 4.0 Ice Cream Sandwich (SDK; first device Galaxy Nexus, Nov 2011). Roboto** (Christian Robertson) replaces Droid Sans as the system font with the Holo design language. Note on sourcing: the Android 4.0 platform announcement post itself does not mention Roboto—the release/date primary is the platform post, while the Roboto attribution rests on the ICS-era Android Design typography page (Wayback snapshot; body not re-read this research) and contemporaneous press. Japanese: **MotoyaLMaru/MotoyaLCedar** (Motoya Co.) bundled, MotoyaLMaru serving as the JP fallback ahead of Droid Sans Fallback (these later became Google Fonts' Kosugi/Kosugi Maru).[111–115]
- **2014-07-16 — Roboto v2 published.** Christian Robertson's redesigned Roboto ("The New Roboto", Google Developers Blog) accompanies Material Design (announced at Google I/O 2014-06-25): rounder forms, new weights. It ships as the system default with **Android 5.0 Lollipop (2014-11-12)**—publication and OS shipment are distinct dated events.[116–118]
- **2017-08-21 — Android 8.0 Oreo. Downloadable Fonts API** and fonts-as-XML-resources (API 26)—apps no longer need to bundle font files; provenance-relevant because app typography stops implying APK contents.[119]
- **Pixel vs AOSP (standing distinction).** Google's proprietary **Product Sans/Google Sans** appears on Pixel devices from Android 8.1 (Dec 2017) in branding surfaces, with **Google Sans Text** spreading through Pixel system apps around Android 12 (2021)—but these are Pixel skin overlays; **the AOSP default remains Roboto throughout**, and **Roboto Flex** (the 12-axis variable font released to Google Fonts on 2022-05-05) has *not* been adopted as the AOSP system default. Pixel-lane version attributions rest on secondary observation and are marked UNCERTAIN.[120–122]

CJK lane.

- **2014-07-15 — Source Han Sans / Noto Sans CJK released.** The first open-source Pan-CJK family—a joint Adobe (Source Han Sans) and Google (Noto Sans CJK) release—the same typeface design under two brands (the released fonts differ in naming tables and packaging)—with partner foundries SinoType (SC), Iwata (JP), and Sandoll (KR); seven weights with distinct SC/TC/JP/KR regional variants.[123, 124]
- **2014-11-12 — Android 5.0 Lollipop. Noto Sans CJK becomes Android's CJK default**, replacing Droid Sans Fallback and delivering per-region glyph correctness for the first time (`fonts.xml` with `zh-Hans`/`zh-Hant`/ `ja`/`ko` language tags; only the Regular weight integrated in practice).[123–125]
- **2015 — Android 6.0 Marshmallow. Legacy Droid Sans Fallback and Motoya files removed** once Noto coverage is complete—corroborated by a system-partition diff of Nexus builds 5.1.1 vs 6.0.0; from Android 7.0 region selection moved to `ttcIndex` entries within a single Noto CJK collection file.[126]
- **2017-04-03 — Source Han Serif / Noto Serif CJK released** (the serif companion). Whether/when stock Android bundled a serif CJK by default is **not established**—no AOSP primary found; treat serif-CJK presence on a device as app- or vendor-supplied absent other evidence. Variable Noto CJK on stock Android is community-attested for **Android 15+** only.[127–129]

3.7 Desktop Linux

Figure 7 traces desktop Linux default UI fonts.

Foundational packages.

- **1999 — Arphic Public License release.** Taiwan's Arphic Technology (文鼎) donates four TrueType fonts—AR PL Mingti2L Big5, AR PL KaitiM Big5 (TC), AR PL SungtiL GB, AR PL KaitiM GB (SC)—under the copyleft **Arphic Public License** (not the GPL, a common miscitation). These are the foundational free Chinese fonts; firefly's bitmap-patched derivatives became the **AR PL UMing/UKai** collections that long served as Debian/Ubuntu Chinese defaults.[130–132]
- **2003-01-22 — Bitstream Vera.** The GNOME Foundation announces the agreement with Bitstream to release the ten-font Vera family (Jim Lyles) under a permissive license—the de facto Linux desktop faces of the mid-2000s.[133]
- **2004 — DejaVu.** Štěpán Roh forks Vera 1.10 to extend coverage; DejaVu becomes the default sans/serif/mono across Fedora, Debian, Ubuntu and openSUSE for roughly 2005–2011 and remains a fontconfig fallback staple (final release 2.37, 2016).[134]
- **2007-05-09 — Liberation fonts.** Red Hat (with Ascender; design Steve Matteson) releases Liberation Sans/Serif/Mono, **metrically identical to Arial, Times New Roman, and Courier New**—"when substituted for the Microsoft fonts, a line of text is identically displayed" per Red Hat's announcement. Relicensed SIL OFL (v2.0, 2012, rebased on Croscore). Default document faces in Fedora and LibreOffice (§4.2).[135, 136]
- **2016-10-06 — Noto unified release.** Google/Monotype's "no more tofu" family publicly launches as a unified whole (components shipped earlier—Android used Noto CJK from 2014, §3.6).[137]

Desktop/distro default switches.

- **2010-10-10 — Ubuntu 10.10.** The **Ubuntu Font Family** (Dalton Maag, funded by Canonical, Ubuntu Font Licence 1.0) becomes the default UI font, replacing DejaVu Sans; Launchpad confirms the 10.10 default installation and authorship.[138]
- **2011-04-06 — GNOME 3.0. Cantarell** (Dave Crossland) becomes GNOME's default UI font—GNOME's own announcement: "the GNOME desktop will use Cantarell by default." Fedora inherits it from Fedora 15 (May 2011) as the Workstation default.[139, 140]
- **2014-07-15 — KDE Plasma 5.0.** The Breeze theme makes **Noto Sans** the Plasma default. Precision note: KDE's own 5.0 announcement (primary for the date and the Breeze theme) does not name the font; the Noto attribution rests on distro configuration records and community documentation, and is labeled accordingly.[141]
- **2022–2023 — Debian's generic aliases move to Noto.** fontconfig 2.14 changed the generic Latin defaults (sans/serif) from DejaVu to Noto in `60-latin.conf`, landing in the Debian bookworm era; after objections Debian reverted the *monospace* alias to DejaVu Sans Mono (fontconfig 2.14.2-4). This is the fontconfig alias layer, distinct from any desktop's UI font.[142]
- **2025-03-19 — GNOME 48. Adwaita Sans**—a customized build of Rasmus Andersson's **Inter**—replaces Cantarell as GNOME's UI font, and **Adwaita Mono** (an Iosevka build) replaces Source Code Pro: the first GNOME UI-font change since 3.0. Fedora 42 ships it via GNOME 48. GNOME's release notes describe the new fonts; the replaced-predecessor framing is corroborated by Phoronix and the GNOME development blog.[143–145]
- **RHEL note.** Red Hat Display/Text (2019, MCKL) are **brand fonts**, not the RHEL desktop UI default—RHEL's GNOME desktop follows the GNOME lineage above. Asserting "RHEL defaults to Red Hat Display" would be incorrect.[146, 147]

CJK lanes on Linux.

- **Japanese.** The free-font lineage runs Kochi (deprecated after the 2003-06-27 glyph-copying disclosure against TypeBank's Mincho-M) → Sazanami (2004, interim) → **VL Gothic** (Fedora's JP default from Fedora 9, 2008) and **IPA fonts** (open-sourced under the OSI-approved IPA Font License, 2009-04-01) → **Takao** forks (Ubuntu JP default from 10.04, first release 2010-02-15) → **Noto Sans CJK JP** (Ubuntu 18.04; Fedora 29).[148–154]
- **Chinese.** Arphic/UMing/UKai (above) → **WenQuanYi Zen Hei** (released 2007-09-15; project founded 2004 by Qianqian Fang), the first open-source Chinese heiti, default for Simplified Chinese in the Fedora/Ubuntu 2007–2008 cycle (exact first-default releases UNCERTAIN) → **Noto Sans CJK** (Ubuntu 16.04 for Chinese—with a famous fontconfig ordering bug that let the JP variant win for zh users until reprioritized; Fedora 29 system-wide).[152–155]
- **Korean.** Baekmuk (1990s) → UnFonts (Ubuntu default ∼2007) → **Nanum** (Naver, released 2008; Ubuntu's Korean default from 12.04) → Noto CJK KR (Fedora 29).[152–154, 156, 157]
- **The Fedora 29 consolidation (2018).** Fedora's Change "CJKDefaultFontsToNoto" (owners Akira Tagoh, Peng Wu) formally switched **all** CJK defaults—VL Gothic (JP), NanumGothic (KR), and the Source Han variants (SC/TC)—to Noto CJK. Negative finding worth recording: the widely-repeated claim that Fedora 21 (2014) made Noto the CJK default has **no primary Fedora**

1999	Arphic donates four Chinese TrueType fonts under the Arphic Public License
2003	Bitstream Vera released (Jan 22, GNOME Foundation agreement); Kochi incident deprecates the free Japanese defaults (Jun 27)
2004	DejaVu forks Vera—becomes the de facto distro default for years
2007	Liberation fonts (May 9, Red Hat and Ascender)—metric-compatible with Arial, Times New Roman, Courier New; WenQuanYi Zen Hei (Sep 15) for Chinese
2010	Ubuntu 10.10 (Oct 10)—Ubuntu Font Family (Dalton Maag) becomes the Ubuntu default
2011	GNOME 3.0 (Apr 6)—Cantarell (Dave Crossland) becomes the GNOME default
2014	KDE Plasma 5.0 (Jul 15)—Breeze theme with Noto Sans as default
2016	Noto unified release (Oct 6); Ubuntu 16.04 switches Chinese to Noto Sans CJK
2018	Fedora 29 switches all CJK defaults to Noto; Ubuntu 18.04 switches Japanese to Noto
2023	Debian bookworm era—fontconfig 2.14 switches generic Latin defaults from DejaVu to Noto
2025	GNOME 48 (Mar 19)—Adwaita Sans (an Inter fork) replaces Cantarell; Adwaita Mono (Iosevka) replaces Source Code Pro

Figure 7: Desktop Linux default UI fonts—foundational packages and desktop switches.

Change behind it; the documented system-wide switch is F29.[152–154]

4 Office Suites

4.1 Microsoft Office

Figure 8 traces the Microsoft Office default document fonts across Latin and CJK edition lanes.

Latin lane.

- **Word for Windows through Word 97:** Normal-template default **Times New Roman 10 pt**; Excel **Arial 10**. **Office 2000 (launch announcement 1999-06-07; retail availability 1999-06-10)** raised Word's default to **Times New Roman 12 pt**—at Word 2000, not 2002/2003 as commonly misremembered—persisting through Word 2003.[158, 159]
- **Office 2007 (volume-license 2006-11-30; retail GA 2007-01-30): Calibri 11 pt** replaces Times New Roman as the Word default and Arial as the Excel/PowerPoint/Outlook default, with Cambria as the serif heading companion—the arrival of the ClearType Font Collection (§3.1). The change was championed by Word program management (Joe Friend) for screen readability. The "Fontgate" affair (2017) turns on the *earlier* of Calibri's two public boundaries: the disputed documents were represented as executed in February 2006, before even the Office 2007 Beta 2 public distribution of 2006-05-23—let alone the 2007-01-30 general availability.[24, 160, 161]
- **Office 2010–2021:** Calibri retained; Office 2013 introduced the Calibri-plus-Calibri-Light theme ("Office Theme 2013–2022").[162]
- **2021-04-28 — "Beyond Calibri":** Microsoft announces five commissioned candidates (Bierstadt, Tenorite, Skeena, Seaford, Grandview), all added to the Microsoft 365 font picker. **2023-07-13:** Bierstadt, renamed **Aptos** (Steve Matteson), is announced as the new default; staged rollout makes it the Microsoft 365 default through late 2023–early 2024, and **Office LTSC 2024 / Office 2024** (2024-09-16 / 2024-10-01) are the first perpetual releases defaulting to Aptos.[32, 33, 163, 164]
- **Cloud fonts:** Microsoft 365's on-demand "cloud fonts" arrived in the 2016–2017 window (exact launch date UNCERTAIN against Microsoft documentation)— provenance-relevant because subscriber documents may use fonts never locally installed.[165]
- **Office for Mac:** Office 2008 for Mac reportedly defaulted to Cambria (a Mac-vs-Windows inconsistency), aligning to Calibri by Office 2016 for Mac—[secondary], with per-app details UNCERTAIN.[166]

CJK edition lanes. Office binds per-script default fonts in its theme (`theme1.xml`, ISO 15924 script tags), so each language edition has its own default lineage:

- **Japanese: MS Mincho** (body) for decades → **Office 2016 for Windows (2015-09-22): Yu Mincho** body with **Yu Gothic** headings (Excel: MS P Gothic → Yu Gothic). The Yu fonts ship in Windows from 8.1 (§3.2). JustSystems' Ichitaro made the equivalent switch in 2021 (§4.4).[167]
- **Simplified Chinese: SimSun 宋体 → DengXian 等线** as the Office 2016 SC-edition default. Precision matters here: the DengXian *font* first shipped in **Windows 10 version 1507 (2015-07-29)** as part of the SC supplemental fonts; **Office 2016** adopted it as the SC default theme font.[168]

- **1989** Word for Windows era—Times New Roman 10 pt (Excel Arial 10)
- **1999** Office 2000 (retail Jun 10)—Word default becomes Times New Roman 12 pt
- **2007** Office 2007 (Jan 30 retail)—Calibri 11 pt replaces Times New Roman in Word and Arial in Excel, PowerPoint, Outlook
- **2013** Office 2013—Calibri body with Calibri Light headings
- **2015** Office 2016 (Sep 22)—Japanese Word body default becomes Yu Mincho (Excel and PowerPoint follow Yu Gothic theme fonts); Simplified Chinese edition switches to DengXian; Traditional Chinese stays PMingLiU
- **2021** Beyond Calibri (Apr 28)—five candidate successors announced
- **2023** Aptos announced as new default (Jul 13); M365 rollout through early 2024
- **2024** Office LTSC 2024 and Office 2024 (Sep 16 and Oct 1)—first perpetual releases defaulting to Aptos

Figure 8: Microsoft Office default document fonts (Latin and CJK edition lanes).

- **Traditional Chinese: PMingLiU 新細明體 — no change found.** In the sources searched (a scoped negative finding on secondary records, §2), the TC edition retained PMingLiU as its default through current Microsoft 365; Microsoft JhengHei (Vista-era ClearType sans, §3.2) is available but is not the document default. A TC document defaulting to PMingLiU alongside an SC document defaulting to DengXian is thus itself a script-edition discriminator.[169]
- **Korean:** Windows' Korean *UI* default moved Gulim → Malgun Gothic at Vista (§3.2), but **which Office release (if any) switched the Korean document body default from Batang-era faces to Malgun Gothic is UNCERTAIN**—no clean primary source found; not asserted.[170]
- **Aptos-era theme bindings (verified by artifact inspection):** Aptos covers Latin/Greek/Cyrillic (plus Vietnamese) only, so the Aptos theme resolves CJK text through per-script `<a:font>` bindings. Direct inspection of the bundled `Office Theme.thmx` → `theme1.xml` in an installed Microsoft Word for Mac (Microsoft 365, version 16.111.1, observed 2026-07-24) shows: **majorFont** (headings) latin = *Aptos Display*, with Jpan = 游ゴシック Light (Yu Gothic Light), Hang = 맑은고딕 (Malgun Gothic), Hans = 等线 Light (DengXian Light), Hant = 新細明體 (PMingLiU); **minorFont** (body) latin = *Aptos*, with Jpan = 游ゴシック, Hang = 맑은고딕, Hans = 等线, Hant = 新細明體. Note the Japanese nuance this exposes: the *theme* body binding is **Yu Gothic**, while Word's Japanese-edition document body default is documented as **Yu Mincho** [secondary]—which implies a template-level override of the theme binding in ja-JP Word (the ja-JP Normal template itself was not inspected here), whereas Excel/PowerPoint follow the theme's Yu Gothic.[171]

4.2 StarOffice, OpenOffice.org, LibreOffice

Figure 9 traces the StarOffice, OpenOffice.org and LibreOffice defaults.

- **StarOffice 5.x (Sun era, 1999–2000).** Bundled Agfa Monotype's metric-compatible clones—**Albany** (≙ Arial), **Thorndale** (≙ Times New Roman), **Cumberland** (≙ Courier New)—so documents exchanged with Microsoft Office kept identical line breaks. The "AMT" suffix on enterprise systems is "Agfa Monotype". Precise first-bundling point release UNCERTAIN; StarOffice 5.2 released 2000-06-20.[172, 173]
- **OpenOffice.org.** 1.0 released 2002-05-01 (announcement 2002-04-30); the proprietary AMT clones could not be redistributed, so OOo leaned on system fonts, with the Writer default carried as "Times New Roman" *by name* (substituted on Linux)—the exact default string is UNCERTAIN against primary documentation. Bundling history: Bitstream Vera (through 2.3) → **Liberation from 2.4** (2008-03) → Gentium added at 3.2 (2010-02).[174, 175]
- **LibreOffice.** Forked 2010-09-28; 3.3 released 2011-01-25. **Writer's default has been Liberation Serif 12 pt from 3.3 to the present** (Impress/Calc: Liberation Sans)—no shipped change found across 3.x–25.x, itself a provenance finding. **LibreOffice 4.1 (2013-07-25) began bundling Carlito and Caladea** (Google's "crosextra" faces, metric-compatible with Calibri and Cambria), the substitution pair that makes post-2007 `.docx` files lay out correctly without Microsoft fonts—a Carlito rendering of Calibri-styled text indicates metric substitution on a host lacking Calibri (LibreOffice bundles Carlito; so do many Linux distributions), not uniquely a LibreOffice host. CJK: LibreOffice's per-locale default lists (`VCL.xcu`) prefer Source Han Sans/Noto Sans CJK first, falling back to platform faces (Microsoft YaHei/NSimSun on Windows, PingFang on macOS), so the effective CJK default

1999	StarOffice 5.x (Sun)—Agfa Monotype metric clones bundled—Albany (Arial), Thorndale (Times New Roman), Cumberland (Courier New)
2002	OpenOffice.org 1.0 (May 1)—relies on system fonts; Writer default carried as Times New Roman by name
2008	OOo 2.4 (Mar)—Liberation fonts bundled
2011	LibreOffice 3.3 (Jan 25)—Writer default is Liberation Serif 12 pt, Impress and Calc use Liberation Sans
2013	LibreOffice 4.1 (Jul 25)—Carlito and Caladea bundled as Calibri and Cambria metric substitutes

Figure 9: StarOffice, OpenOffice.org and LibreOffice defaults.

is host-dependent.[176–181]

4.3 WPS Office, Google Docs, Apple iWork

- **WPS Office (Kingsoft).** For Chinese documents WPS follows the SimSun convention—behavioral testing shows 宋体 (SimSun) as the effective Chinese default and missing-font fallback (STSong on macOS), matching Chinese-locale Word; its Latin default is UNCERTAIN against vendor documentation. WPS has integrated third-party free fonts (e.g., Xiaomi's MiSans) rather than shipping a proprietary clone set. It is widely deployed in China, including a dedicated government "Official Document Edition" (2022); market-share figures circulating for WPS (e.g., a ∼95%-by-1993 claim) survive only in tertiary compilations and are not asserted here.[182, 183]
- **Google Docs.** Default body font **Arial 11** with 1.15 line spacing, as documented for recent Google Docs; no record of a default-face change was found in the sources searched, but the earliest-years (2006-era) default is unverified—the continuous-since-2006 reading is UNCERTAIN. Google Fonts (launched May 2010) later fed Docs' "more fonts" picker, expanding available—not default—fonts.[184, 185]
- **Apple iWork Pages.** Recent Pages versions default the Blank template body to **Helvetica Neue 11 pt** per secondary documentation (Pages has no global default font—templates' paragraph styles carry it); this was not version-verified against a named Pages build during this research. Pre-2013 template defaults (Pages '05–'09) are UNCERTAIN against Apple primary documentation and are not asserted here.[186, 187]
- **Historical one-liners, explicitly unverified.** WordPerfect for Windows (commonly reported Times New Roman 12, earlier versions Courier) and Lotus Word Pro (commonly reported Times New Roman)—both UNCERTAIN; presented only as commonly reported.[188]

4.4 CJK-market office suites: Ichitaro and Hancom

- **JustSystems Ichitaro 一太郎 (Japan; first released 1985; the #2 Japanese word processor after Word).** Default Japanese body font was **MS Mincho through Ichitaro 2020**; per JustSystems' support note #058017 and contemporaneous trade-press reporting, **Ichitaro 2021 (released 2021-02-05) switched the default to Yu Mincho 游明朝**—mirroring Office 2016's JP switch (§4.1) five years later (the vendor note itself was not fetched during this research; attribution rests on the reporting). Premium editions bundle licensed professional fonts (Morisawa faces, Chikushi 筑紫 families, UD fonts).[189–191]
- **Hancom Office / Hangul HWP (Korea; dominant in Korean government).** The historical Korean default serif category is **바탕 Batang** (the 1993 Ministry of Culture standardization renamed the category from *myeongjo*); recent Hancom versions are reported to default to Hancom's own **함초롬바탕 HCR Batang** [secondary — UNCERTAIN] (Hamchorom family; only HCR Batang/Dotum are licensed for use outside Hancom Office). The exact HWP release that switched to Hamchorom is UNCERTAIN (Hangul 2010/2014 era).[192, 193]

5 PDF and Scan/OCR Tools

5.1 Adobe Acrobat and Reader

Figure 10 traces the Adobe Acrobat font-substitution provenance markers.

- **1993-06-15 — the Base-14.** PDF 1.0 (announced at Comdex Fall 1992; Acrobat 1.0 shipped June 1993) guarantees **14 standard Type 1 fonts** every conforming viewer supplies without embedding: Times, Helvetica, and Courier in four styles each, plus Symbol and ZapfDingbats (normative in the PDF Reference and ISO 32000-1 §9.6.2.2). Acrobat on Windows maps them to metrically compatible system faces (Helvetica→Arial, Times→Times New Roman, Courier→Courier New); other viewers ship or select their own metrically compatible substitutes.

- **1993** Acrobat 1.0 and PDF 1.0 (Jun 15)—Base-14 standard fonts guaranteed without embedding
- **1994** Acrobat 2.0—Adobe Sans MM and Adobe Serif MM multiple-master substitution fonts introduced
- **2006** Acrobat 8 era—Typewriter tool defaults to Courier; form fields Helvetica 12
- **2011** Acrobat X and later—Edit PDF and TouchUp fall back to Minion Pro for missing Latin fonts

Figure 10: Adobe Acrobat font-substitution provenance markers.

Provenance: unembedded Base-14 face names mark programmatically-generated or default-styled PDFs relying on viewer substitution.[194–198]

- **1994 — Acrobat 2.0: Adobe Sans MM / Adobe Serif MM.** Two multiple-master fonts distributed only inside Acrobat/Reader (not installable or licensable separately), used to synthesize metric-matched stand-ins for missing non-Base-14 Latin fonts. *Provenance:* their appearance in rendered or re-saved output is strongly consistent with Acrobat processing a PDF whose fonts were absent—the strength of the inference comes from their Acrobat-only distribution, though other Adobe-derived workflows can in principle carry the names.[199, 200]
- **Tool defaults.** Typewriter tool: **Courier** (Acrobat 8 era), later Helvetica; form text fields: **Helvetica 12**; Fill & Sign: fixed by design. Edit PDF/TouchUp fallback for unavailable Latin fonts: **Minion Pro** (Acrobat X/XI/DC; configurable via Content Editing preferences). *Provenance:* Minion Pro body text inside an otherwise foreign document is consistent with post-hoc Acrobat editing— corroborate via Acrobat metadata or incremental-update history, since Minion Pro is a widely distributed Adobe typeface in its own right.[201–204]
- **CJK substitution packs.** Acrobat's Asian font packs supply **Kozuka Mincho/Gothic** (JP), **Adobe Song Std** (SC), **Adobe Ming Std** (TC), **Adobe Myungjo Std** (KR) as substitution faces for unembedded CJK fonts (pack composition evolved across Acrobat 7→DC; the per-version, per-lane pack composition is reported from vendor and user documentation and is UNCERTAIN at per-version level). *Provenance:* unembedded CJK text rendering in these faces is consistent with Acrobat substitution; the same families are used in ordinary typesetting, so corroborate with embedding status and document metadata.[205–207]

5.2 Scan/OCR tools (PaperPort and peers)

Provenance principle: **an OCR'd document's text layer carries the OCR engine's output-font choices, not the original's typography.**

- **ABBYY FineReader** (best documented): assigns recognized text to visually-closest fonts by default and supports forcing a single output font—commonly Arial or Times New Roman; ABBYY's Linux engine documentation expects Arial/Times New Roman present. A uniform Arial/Times text layer over a scan is consistent with ABBYY-class OCR output.[208–210]
- **PaperPort** (Visioneer 1992 → ScanSoft 1999 → Nuance 2005 → Kofax 2018 → Tungsten Automation 2024): distributed at scale as SOHO scanner bundleware—licensed with Brother, Canon, Dell, HP, and Xerox devices per the bundling model described in Nuance's SEC filings (which evidence the *distribution channel*; no installed-base or market-share figures were located, so "popularity" is not quantified here). Historically carried the OmniPage OCR engine. Its OCR output-font behavior is **undocumented**—stated as such, not invented.[211–213]
- **OmniPage** (Caere, late 1980s → ScanSoft 2000 → Nuance → Tungsten) and **Readiris** (I.R.I.S., Belgium, 1987+, now a Canon company): same class; output-font defaults undocumented—UNCERTAIN.[214, 215]
- **Other PDF editors:** Foxit's add-text defaults to Helvetica; PDF-XChange's Typewriter historically Courier New 12; Nitro exposes no documented single default (UNCERTAIN).[216–218]

6 Document-Generation Libraries (Aspose and Peers)

Documents produced or converted by server-side libraries carry the library's default and substitution fonts—a distinctive toolchain fingerprint.

Key documented facts: Aspose.Words' `DefaultFontSubstitutionRule.DefaultFontName` documents "The default value is 'Times New Roman' "; the substitution order (exact match → embedded → AltName → rules → first available → bundled Fanwood) is Aspose's own documentation, including the characteristic warning "Font 'Times New Roman' has not been found. Using 'Fanwood' font instead."; Aspose.Slides' documentation states Aspose distributes no fonts and picks the closest system font. None of the products ship CJK fonts—CJK output on bare

Table 1: Default and missing-font behavior of the Aspose document-generation suites.

Product	New-object default	Missing-font behavior	Provenance tell
Aspose.Words	Times New Roman 12 (Normal style)	Substitution chain ends at `DefaultFontName` = "Times New Roman"; then first available font; last resort: embedded **Fanwood.ttf**	Fanwood output, with Aspose's substitution warning or matching metadata, is strong evidence of Aspose.Words on a font-starved host (Fanwood alone is only consistent with it—the font is publicly available)
Aspose.Cells	Calibri 11 (reported — UNCERTAIN as cited, see note)	Falls back to workbook `DefaultStyle.Font`; PDF/image save options can override	Calibri mirrors Excel 2007+
Aspose.Slides	none fixed (deck theme fonts)	Heuristic closest-available system font, PowerPoint-like; ships no fonts	No single fingerprint; Cambria Math irreplaceable for equations
Aspose.PDF	Times New Roman (reported — UNCERTAIN as cited, see note)	Exact-name match, then `FontRepository.Substitutions`; Base-14 never embedded	Times New Roman re-set on edited text

containers produces tofu or host-dependent fallback (the ordering is *not* Aspose-defined), and Aspose.PDF throws on standard CJK PostScript names (STSong-Light, MSung-Light, HYSMyeongJo-Medium) when absent. **Stability finding (scoped):** in the Aspose release notes and documentation searched (July 2026, keyed on default-font changes), no change to these defaults was found—in particular nothing tracking Microsoft's 2024 Aptos switch. Within that search scope the defaults appear stable, which cuts both ways: an Aspose default alone cannot narrow a document to a release window, and an Aptos-defaulted document is inconsistent with a factory-default Aspose path.[219–224]

Peers, for contrast—only openpyxl is source-verified here; the others are commonly documented, version- and API-dependent, and UNCERTAIN as cited: **openpyxl** hardcodes Calibri 11 (`DEFAULT_FONT` in `styles/fonts.py`); **iText**'s text APIs are documented as defaulting to Helvetica; **Apache PDFBox** has no implicit default (the caller chooses); **python-docx** inherits Calibri 11 from its bundled template; **GemBox** documentation reports Calibri defaults.[225]

7 Cross-Platform Provenance Anchors (Quick Reference)

Selected anchors, each typed—because "first design", "first public beta", "first OS bundling", and "first default binding" are different events with different evidential force. Anachronism reasoning must use the **earliest public availability of any kind, including public betas and standalone font releases**; the later anchor types (bundling, default binding) support consistency reasoning about producing toolchains, not impossibility claims. See the per-section registers for citations (Table 2).

8 Limitations

1. **Verification asymmetry.** Every citation carries its verification tier; [secondary] and UNCERTAIN labels are load-bearing. The residual weak points are enumerated in place: pre-2000 CJK regional-edition ship dates (Microsoft's records begin at Windows 2000); classic-Mac-era dates; the SF Pro rebrand year (2017, pending a Wayback body read); several Wayback snapshot bodies confirmed to exist but unread due to archive.org rate limiting during the research window; the explicitly unverified legacy office defaults (§4.3); and the Aspose.Cells/Aspose.PDF and iText/python-docx/GemBox default claims, whose specific source pages were not captured and which are marked UNCERTAIN in their footnotes.

2. **Default-font attribution has no base rates.** No empirical evidence is offered here for how often users, organizations, templates, or converters retain factory defaults. Default-font matches therefore support only *consistent-with* statements about candidate toolchains—never probabilistic attribution, and phrases elsewhere in this paper are calibrated accordingly. Anchor dates give *terminus post quem* logic only in the direction "font used before its earliest public availability ⇒ anachronism"; the converse direction is not symmetric.

3. **Availability precedes release for insiders.** Public-release anchors bound *ordinary* acquisition. Designers, foundries, beta testers, OEMs, and licensees hold fonts earlier; where public betas materially widen early access (Calibri), the beta date is the operative anchor, and §7 types each anchor for this reason.

4. **Presence ≠ bundling** (modern OSes). Windows 10+ Feature-on-Demand, macOS download-on-

demand, iOS 13+ user font installation, and Android's Downloadable Fonts all decouple a font's presence on a device from the OS release that "shipped" it.

5. **Single-artifact inspections generalize narrowly.** The Office-theme inspection (§4.1) is documented with path, version, date, and SHA-256, and is valid for that build and channel only.

6. **Scope.** Timelines cover public/general releases; insider builds, betas, and enterprise LTSC variants are noted only where they matter (Office 2007 Beta 2, iOS 7 beta 3). "Current" statements are bounded by the July 2026 research window.

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Table 2: Cross-platform provenance anchors, each typed by anchor kind.

Font	Date	Anchor type	Vehicle
Arial, Times New Roman, Courier New (TrueType)	1992-04-06	first OS bundling	Windows 3.1
Comic Sans MS	1995	first bundling (add-on pack/OEM)	Windows 95 Plus!
Verdana, Georgia	1996	font release (IE/web-fonts program); OS-bundled by 98/2000	Microsoft web fonts
Tahoma	1994–1995 (Office/IE); 2000-02-17	font release; then first OS default binding	Windows 2000
Lucida Grande	2000-09-13	first OS bundling and default (public beta)	Mac OS X Public Beta
Hiragino (on Mac)	2001-03-24	first Mac OS bundling (existed earlier as a Jiyukobo/SCREEN commercial family)	Mac OS X 10.0
Segoe UI	2006-11-30 / 2007-01-30	first OS bundling and default	Windows Vista
Calibri and the ClearType six	2006-05-23; GA 2007-01-30	first public availability (Office 2007 Beta 2); then GA	Office 2007 / Vista
Microsoft YaHei, JhengHei, Meiryo, Malgun Gothic	2007-01-30	first OS bundling and per-locale default	Windows Vista
Droid Sans	SDK 2008-09-23; first shipping handset Oct 2008 (announced 2007-11-12)	first OS bundling and default	Android 1.0 / T-Mobile G1
Ubuntu Font Family	2010-10-10	first OS bundling and default	Ubuntu 10.10
Cantarell	2011-04-06	first desktop default binding (font designed 2009)	GNOME 3.0
Roboto	2011-10-19 (SDK) / Nov 2011 (device)	first OS bundling and default	Android 4.0
Yu Gothic / Yu Mincho	2013-10-17	first OS bundling (existed earlier as Jiyukobo commercial faces)	Windows 8.1
Source Han Sans / Noto Sans CJK	2014-07-15	font release (open source)	Adobe/Google
San Francisco (SF)	2014-11-18 (developer) / 2015-04-24 (watch) / Sep 2015 (iOS 9, El Capitan)	developer release; then OS default	Apple platforms
PingFang SC/TC/HK	2015-09-16 (iOS 9) / 2015-09-30 (El Capitan)	first OS bundling and default	Apple platforms
DengXian 等线	2015-07-29	first OS availability (Windows 10 FOD); Office 2016 SC default binding later	Microsoft
Segoe UI Variable, Cascadia Code	2021-10-05	first OS bundling (Cascadia Code open-sourced 2019)	Windows 11
Roboto Flex	2022-05-05	font release (never an AOSP default)	Google Fonts
Aptos	announced 2023-07-13; rollout late 2023–early 2024	app default announcement, then staged rollout (M365 cloud fonts)	Microsoft 365
Adwaita Sans	2025-03-19	first desktop default binding (Inter, its basis, released 2016)	GNOME 48